

Thinking Critically and Analytically about Critical-Analytic Thinking: an Introduction

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In October 2013, 18 scholars from such varied disciplines as educational policy and leadership, cognitive neuroscience, developmental psychology, language and literacy, social psychology, measurement and assessment, and educational psychology gathered together for 2 days of intense and informative discussion and debate. The catalyst for this gathering was a grant from the American Educational Research Association's Education Research Conferences Program, awarded to Alexander, Wentzel, and Dunbar, intended to stimulate interdisciplinary discourse around complex and significant educational issues with the goal of promoting research initiatives and sparking reform in policy and practice. In keeping with that laudable intention, the goal of this specific assembly of multidisciplinary scholars was to tackle one of the more salient and recurring themes in contemporary educational theory and practice: What does it mean to think critically and analytically and how can critical-analytic thinking be measured and fostered in our children and youth?

The purpose of this special issue on critical-analytic thinking is to share conundrums, insights, and recommendations that arose from this multidisciplinary assembly and the subsequent examinations that have arisen from the continuing conversations among the attending scholars with the broader educational psychology community. The cross-disciplinary contributions that form this special issue deal with such core topics as the nature, development, assessment, and training of critical-analytic thinking. Those who have contributed to this conference and this issue feel strongly that these scholarly treatises can prove invaluable for those already invested in or concerned with this foundational subject, as well as those seeking to better understand what critical-analytic thinking is, why it matters, and how it can be measured or shaped.

There is no question that changing standards for curriculum, as evidenced in the Common Core State Standards (CCSS; National Governors Association Center for Best Practices and Council of Chief State School Officers 2010), now having been adopted by 48 states, and the Next Generation Science Standards (NGSS Lead States 2013), seeking to transform science practices, have put the construct of critical-analytic thinking in the forefront. Nonetheless,

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despite the coordinated and aggressive efforts to build the assessment tools needed to gauge advancements along these new curricular fronts, including Smarter Balanced Assessment Consortium (Smarter Balanced Assessment 2013) and the Partnership for Assessment of Readiness for College and Careers (Partnership for Assessment of Readiness for College and Careers 2014), it is readily apparent that the very construct of critical-analytic thinking remains elusive, especially when the broader and diverse landscape of education research is surveyed. There is no definitive definition of critical-analytic thinking that shapes current policies and practices and no highly regarded or widely accepted measure of critical-analytic thinking that exists within the field—at least not yet.

In effect, although such significant curricular and assessment initiatives as CCSS or PARC speak clearly and directly to the need for school-aged populations to engage in critical-analytic thinking, there appears to be little in the way of consensus or certainly within the extant literature regarding critical-analytic thinking's nature and the course of its development. Likewise, there are serious concerns as to how the individual differences that humans manifest may alter the process and outcomes associated with critical-analytic thinking. Further, there are significant issues pertaining to how this elusive construct can be and should be reliably and validly assessed or what techniques or approaches work best to promote critical-thinking in children and youth—both in and out of school. Thus, this special issue serves an important and timely purpose for the educational research community and for those in positions of educational leadership who are charged with the improvement of the educational experience for all learners.

Critical-Analytic Thinking

One long-held belief shared by educational researchers, practitioners, policymakers, and the general populace is that schools and society benefit when individuals or groups are perceptive and attentive to the world around them and manifest the ability to think deeply and flexibly about important issues, that is, when they think critically and analytically (e.g., Dewey 1933; Bloom 1956; Halpern 2001; Ku 2009). The very foundation of a democratic society is a citizenry capable of and willing to engage in such critical-analytic thought (Westheimer 2008). Moreover, education within democratic societies is presumed to contribute to the development of such a thinking and reasoning populace. However, there is ample evidence that this laudable goal is unrealized for many citizens of the USA and that far too many students lack the knowledge, beliefs, skills, and strategies required to think critically and analytically (Arum and Roksa 2011).

Although investment in the study of critical-analytic thinking has waxed and waned over the past century, there is currently a heightened interest in this higher-order mode of thought, as well as an increased concern as to how to promote it in students of all ages, backgrounds, and abilities (Schraw and Robinson 2012). There are many reasons that interest in critical-analytic thinking has moved to the forefront in the scholarly discourse, and its investigation has been approached from a variety of theoretical and methodological perspectives. For one, the prevalence of high-stakes testing appears to reinforce beliefs about knowledge and knowing (i.e., epistemic beliefs) that favor overly simplistic and narrow views of knowledge and a preference for quick and unjustified responses (Alexander et al. 2011; Alexander and The disciplined reading and learning research laboratory 2012). The growing interest in critical-analytic thinking or other forms of higher-order thinking or reflective reasoning is, thus, seen as desirable alternatives to the “dumming down” of the educational experience and the potential threat such oversimplification poses to human potential.

For another, there is a burgeoning body of cognitive and neuroscience research that amplifies the importance of fostering critical-analytic forms of thinking within developing children and youth (Mackey et al. 2011), and the deleterious effects of poverty on brain functioning and critical thinking (Farah 2010). Through advancements in the study of human variability, in general, and the brain and its functioning; in particular, we are now better positioned to understand what critical-analytic thinking may cognitively and neurologically entail, how malleable the components of critical-analytic thinking may be, and when those components manifest both in typically and atypically developing individuals.

In addition, the social developmental literature suggests that the quality of interactions with parents, teachers, and peers can have a significant impact on the development of higher-order cognitive skills, including critical-analytic thinking (Case 2005; Dunbar and Klahr 2012; Wentzel 2009a). We witness those benefits in home environments where parents encourage even young children to explore and to question the world around them and to plant the seeds of literacy and numeracy within those enriched interactions (Ramani et al. 2010; Rowe 2008). Similarly, there is growing evidence that infusing classrooms with quality discourse that not only guides children and youth to weigh alternatives and seek best evidence for the choices and judgments they make (Murphy et al. 2009) but also can be a significant path toward improved critical-analytic thinking.

Similarly, the sociocultural climates of classrooms and schools have been shown to matter greatly in the frequency and quality of higher-order modes of thinking (Bassok et al. 2012; Halpern 2001; Holyoak 2012). Research on the development of motivation to learn also suggests that the nature of teacher-student interactions can influence the thought processes that students lend to solving academic tasks (Molden and Miele 2008; Ramani et al. 2010; Wentzel 2009b; Wentzel and Wigfield 2009; Wigfield and Guthrie 2010). Critical-analytic thinking cannot be solely regarded as a “coldly cognitive” endeavor (Pintrich et al. 1993). Individuals’ goals and perspectives on learning and knowing, their conceptions about their own capabilities, and the beliefs they hold about the malleability of intelligence have all been shown to matter—to matter in the efforts made to reason and reflect and the outcomes reached. What further exacerbates concerns over critical-analytic thinking is that the focus on basic skills within identified populations, such as struggling readers or second language learners (Bolger et al. 2008; Wang 2011), may restrict their opportunities to engage in critical-analytic thought or to acquire the knowledge, beliefs, and strategies that are foundational to this manner of thinking and reasoning. In effect, the learning environment for these identified populations may increase the likelihood that struggling readers or writers become struggling thinkers. Thus, the more that we come to understand the

1. Interplay among the nature and development of critical-analytic thinking
2. Influence of learner cognitive, neurological, linguistic, and social-motivational differences
3. Effects of assessment practices to shape and to gauge critical-analytic thinking; and
4. Characteristics of in-school and at-home environments including the quality of interactions that transpire between parents, teachers, or peers and the developing minds of children and youth who populate those environments,

the more educators, policymakers, and a concerned populace will achieve the understanding of critical-analytic thinking they seek.

Common Ground Claims and Model

With this general framing as backdrop, the scholars in attendance at the invitational conference, “Seeking Common Ground,” set out to engage in critical-analytic thinking about critical-analytic thinking. Based on their empirical and theoretical knowledge rooted in varied disciplines, these scholars participated in rich discussions and rigorous debate—posing arguments and counterarguments supported by competing perspectives and research traditions. Yet, as a collective, they shared the espoused goal of not only solely sharpening the differences that existed among them but also of striving for consilience, a term introduced by Whewell (1840) and used by Wilson (1998) to describe unity of knowledge across varied disciplines.

For the proposers of this conference (Alexander, Wentzel, and Dunbar), this consilience would hopefully take the form of shared claims about critical-analytic thinking that crossed philosophical, psychological, and methodological lines and that could guide future research and instructional practice in the years to come. The conference organizers did not intend that there be complete or unwavering agreement from contributing scholars since such outcomes are rarely realized in interdisciplinary examinations of complex constructs. But, they did hope for strong cross-disciplinary statements that could be shared with others. Indeed, this hoped-for goal was realized in the form of the Shared claims that appear in Table 1.

Further, initiated by the thinking of Robert Croninger, a model of the critical-analytic process was initially put forward and subsequently sharpened and supported by the cross-disciplinary contingent. This model appears in Fig. 1 and captures relations among elements noted within the Shared claims. Because these Shared claims and this process model represented interdisciplinary efforts, I offer them here as evidence of the level of consilience that was realized and the productive collaborations that have continued.

Table 1 Common ground on the nature, development, assessment, and teaching of critical-analytic thinking

Shared claims

- Critical-analytic thinking is based on credible evidence.
 - What constitutes credible evidence in critical-analytic thinking varies by task, context, and normative constraints.
 - Critical-analytic thinking is effortful.
 - Critical-analytic thinking can be potentially costly or beneficial.
 - Young children can engage in critical-analytic thinking under the right conditions.
 - Critical-analytic thinking is fundamental to expertise development in cognitive domains.
 - Critical-analytic thinking is important to support the social good.
 - Critical-analytic thinking is fostered by the interplay of cognitive, social, motivational, neurophysiological, or experiential conditions.
 - It is possible to construct learning environments that nurture critical-analytic thinking.
 - Critical-analytic thinking can be individual and collective.
 - The process of critical-analytic thinking should be systematically incorporated into the school experience for all learners.
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THE CRITICAL-ANALYTIC THINKING PROCESS MODEL

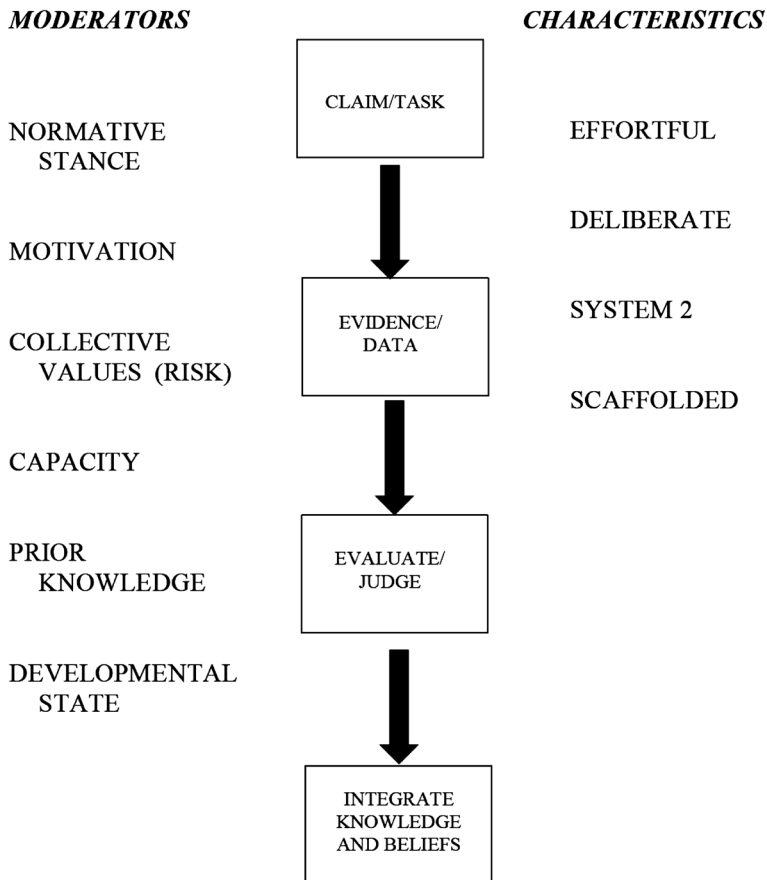


Fig. 1 The critical-analytic thinking process model

Overview of Special Issue Contributions

In developing the individual articles for this special issue, several guidelines were followed. First, each of the articles represents a collaborative and multidisciplinary contribution. For example, in the initial piece by Byrnes and Dunbar (2014), a developmental psychologist and a cognitive scientist take on the subject of the nature of critical-analytic thinking and its development over time and experience. Similarly, in the proposed article by Bolger, Mackey, Wang, and Grigorenko (2014), issues of individual differences in critical-analytic thinking are explored by scholars with backgrounds in counseling and educational psychology, cognitive neuroscience, and second language learning. Further, although Miele and Wigfield (2014) share a deep concern over human motivation, their orientations toward that concern, as social and educational psychologists, follow slightly different paths.

The combined efforts of Brown et al. 2014—whose arenas of expertise are strongly rooted in science, literacy, and educational policy, respectively—serve to give the topic of the assessment of critical-analytic thinking a dimensionality rarely seen. Moreover, with the interdisciplinary perspective afforded by Murphy, Rowe, Ramani, and Silverman, we are able to grasp the possibilities of enhancing critical-analytic thinking in both reading and mathematics, for both younger and older students, and both in home and at school. Their own research efforts, and that of others that they cite, give us renewed hope that positive changes in critical-analytic thinking are indeed achievable.

Second, it was expected that each article would not only afford the readership of *Educational Psychology Review* a concise yet informative overview but would also draw upon varied literatures to establish relevant issues and offer sound recommendations. For instance, the consideration of assessment of critical-analytic thinking offered by Brown et al. (2014) centers on high-stakes testing in the domains of mathematics, science, and literacy and on national policy initiatives, such as the CCSS and the NGSS. Similarly, the contribution of Miele and Wigfield (2014) weaves together the educational and social psychology literatures creatively to consider the relations between motivation and critical-analytic thinking both quantitatively and qualitatively. Further, the article by Bolger et al. (2014) offers a rare glimpse of critical-analytic thinking through a neurocognitive lens that also acknowledges such experiential influences as socioeconomic status and language background.

Third, it was agreed upon that each contribution would tie into one or more of the shared claims about critical-analytic thinking or to the model of the process of critical-analytic thought. Although the contributors to the special issue make this connection in their own unique way, the two articles chosen to bookend this special issue are particularly important in this regard. Specifically, the article by Byrnes and Dunbar (2014) that opens the issue extends our understanding of the nature and development of critical-analytic thinking and gives greater meaning to the attributes that populate the Shared claims (Table 1). Further, what exists as a general model in Fig. 1 is skillfully enacted by Murphy et al. (2014) in their rich description of approaches and guidelines for promoting critical-analytic thinking not just in classrooms but out of school as well. In combination, these two articles help to bring greater clarity to the notion of critical-analytic thinking and serve to demonstrate that critical-analytic thinking is not only possible but also probable when environments are effectively orchestrated with that end in mind.

As a conclusion to the special issue, I have called upon two renowned scholars to provide concluding reflections, Kathryn Wentzel and Daniel Robinson. As one of the initiators of the initial conference that was the catalyst for this special issue, Wentzel offers a critical analysis from the vantage point of a social and educational psychologist, whose research in peer and peer-teacher relationships introduces an important dimension to the discussion critical-analytic thinking. Robinson, a long-time editor of *Educational Psychology Review* and coeditor of a groundbreaking volume on higher-order thinking (Schraw and Robinson 2012), provides his own unique perspective on the topic of critical-analytic thinking and the specific contributions in this issue.

Concluding Thoughts

It is not often that an educational psychologist such as myself can speak with certainty or conviction about the elusive constructs that are the crux of our professional lives. Yet, I feel quite certain that the surge of interest in critical-analytic thinking that is making its way into educational policy and practice needs to be well framed by those with the theoretical

background and empirical wherewithal to do so. The collection of scholars who have shared their insights and recommendations in the pages of this special issue are among those capable of achieving this laudable end. Moreover, the interdisciplinary perspectives they bring to bear on the topic of critical-analytic thinking open new vistas unavailable from more constricted vantage points. Of course, it remains to be seen what comes next. How will the insights and recommendations that can populate this issue translate into empirical studies, educational policies, or curricular resources that ultimately make a difference in the lives of all learners? For now, I can only wait and hope that the outcomes of this joint effort will be positive.

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